

B.Tech. Degree VII Semester Regular/Supplementary Examination November 2023

ME 19-205-0702 VIBRATION AND NOISE CONTROL (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Model a given physical system using spring, mass, and damping elements.
 CO2: Formulate the governing differential equations of single dof, multi dof, and continuous systems.
 CO3: Solve the governing equations and interpret the results.
 CO4: Understand the effects of noise and the control measures adopted.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate,
 L6 – Create
 PO – Programme Outcome

PART A

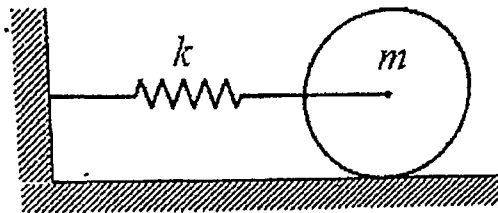
(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	A spring mass system has a time period of 0.25 seconds. What will be the new time period if the stiffness of the spring is increased by 50%?	3	L2	3	1,2,3
(b)	In what ways is the response of a system with Coulomb damping different from that of systems with viscous damping?	3	L3	1	1,2,3
(c)	Derive an expression for the length of a torsionally equivalent shaft.	3	L2	1	1,2,3
(d)	What is the significance of eigen values and eigen vectors in a multi-degree of freedom vibrating system?	3	L1	1	1,2,3
(e)	Derive an expression for the whirling speed of a shaft.	3	L2	2	1,2,3
(f)	Derive the wave equation for longitudinal vibration of bars.	3	L2	2	1,2,3
(g)	Discuss temporary and permanent hearing loss from continuous noise.	3	L1	4	1,2
(h)	Differentiate between sound pressure level and sound intensity level.	3	L1	4	1,2

PART B

(4 × 12 = 48)

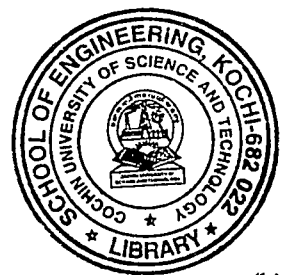
- II. (a) A circular cylinder of mass m and radius r is connected by a spring of stiffness k as shown in figure. If it is free to roll on a rough horizontal surface without slipping, find its natural frequency.



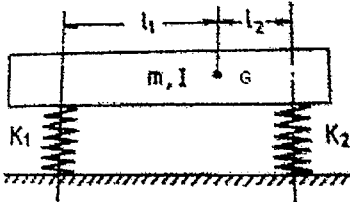
- (b) A machine part having a mass of 2.5 kg vibrates in a viscous medium. A harmonic exciting force of 30 N acts on the part and causes a resonant amplitude of 14 mm with a time period of 0.22 second. Find the damping coefficient. If the frequency of the exciting force is changed to 4 Hz, determine the increase in amplitude of the forced vibrations upon the removal of the damper.

OR

(P.T.O.)



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		Marks	BL	CO	PO
III.	(a) Prove that the ratio of successive amplitudes in an under damped vibrating system is a constant.	4	L2	2	1,2,3
	(b) A body of 5 kg is supported on a spring of stiffness 200 N/m and has dashpot connected to it which produces a resistance of 0.002 N at a velocity of 1 cm/s. In what ratio will the amplitude of vibration be reduced after 5 cycles?	8	L4	3	1,2,3
IV.	A two degree of freedom model of an automobile for coupled translatory and rotary motion in vertical plane is shown in figure. Determine the natural frequencies and the mode shapes. Assume following data for the automobile: $m = 1200$ kg, $I = 1500$ kgm ² , $K_1 = 55$ kN/m, $K_2 = 60$ kN/m, $l_1 = 1.35$ m and $l_2 = 1.15$ m. Assume that the displacement is given with respect to the centre of gravity G .	12	L4	3	1,2,3
					
OR					
V.	A diesel engine flywheel rotating at 250 rpm drives a centrifugal pump at 1000 rpm through a pair of gear and pinion. The moments of inertia of the engine flywheel and the pump impeller are 1400 kg.m ² and 400 kg.m ² respectively. The shaft connected to the engine is 40 mm in diameter and 180 mm long. The pump shaft is 60 mm in diameter and 450 mm long. Determine the natural frequency of torsional oscillations of the system and also the position of node, neglecting the inertia of gear and pinion. The modulus of rigidity for the shaft material is 84 GN/m ² .	12	L4	3	1,2,3
VI.	A shaft of diameter 6 cm and length 5 m is simply supported at the ends and carries four masses 50 kg each at equal distance over the length of the shaft. Determine the whirling speed of the shaft. The weight of the shaft is negligible. Assume modulus of elasticity $E = 200$ GN/m ² .	12	L4	3	1,2,3
OR					
VII.	A uniform beam fixed at one end and simply supported at the other is vibrating in the transverse direction. Derive the frequency equation.	12	L3	2	1,2,3
VIII.	(a) Explain the physiological and psychological effects of noise.	4	L1	4	1,2
	(b) Show that the sound intensity level decreases by 3 dB for cylindrical acoustic waves, as the distance from a point source is doubled.	8	L2	4	1,2
OR					
IX.	(a) Explain the methods adopted for the control of noise at the source and the receiver.	4	L1	4	1,2
	(b) How will you determine the overall sound pressure level when two machines are working simultaneously?	8	L3	4	1,2

Bloom's Taxonomy Levels

L1 = 14.17%, L2 = 20.0 %, L3 = 29.17%, L4 = 36.67%.
